

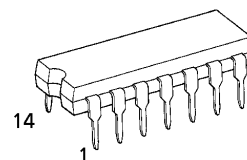
TC4011BP, TC4011BF, TC4011BFN, TC4011BFT

TC4011B QUAD 2 INPUT NAND GATE

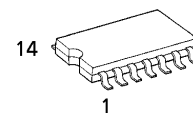
The TC4011B is 2-input positive logic NAND gate respectively.

Since all the outputs of these gates are provided with the inverters as buffers, the input/output characteristics have been improved and the variation of propagation delay time due to the increase in load capacity is kept down to the minimum.

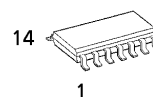
(Note) The JEDEC SOP (FN) is not available in Japan.



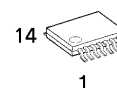
P (DIP14-P-300-2.54)
Weight : 0.96g (Typ.)



F (SOP14-P-300-1.27)
Weight : 0.18g (Typ.)



FN (SOL14-P-150-1.27)
Weight : 0.12g (Typ.)

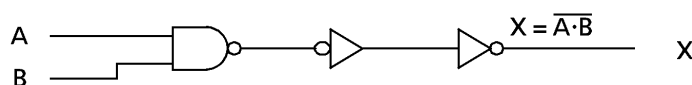


FT (TSSOP14-P-0044-0.65)
Weight : 0.06g (Typ.)

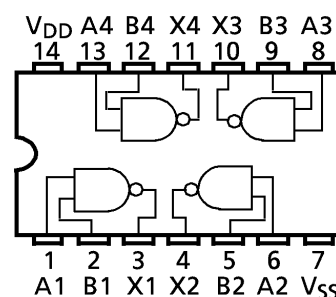
MAXIMUM RATINGS

CHARACTERISTIC	SYMBOL	RATING	UNIT
DC Supply Voltage	V_{DD}	$V_{SS} - 0.5 \sim V_{SS} + 20$	V
Input Voltage	V_{IN}	$V_{SS} - 0.5 \sim V_{DD} + 0.5$	V
Output Voltage	V_{OUT}	$V_{SS} - 0.5 \sim V_{DD} + 0.5$	V
DC Input Current	I_{IN}	± 10	mA
Power Dissipation	P_D	300 (DIP) / 180 (SOIC)	mW
Operating Temperature Range	T_{opr}	$-40 \sim 85$	$^{\circ}\text{C}$
Storage Temperature Range	T_{stg}	$-65 \sim 150$	$^{\circ}\text{C}$

LOGIC DIAGRAM



PIN ASSIGNMENT (TOP VIEW)



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RECOMMENDED OPERATING CONDITIONS ($V_{SS} = 0V$)

CHARACTERISTIC	SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
DC Supply Voltage	V_{DD}		3	—	18	V
Input Voltage	V_{IN}		0	—	V_{DD}	V

STATIC ELECTRICAL CHARACTERISTICS ($V_{SS} = 0V$)

Characteristic		Sym- bol	Test Condition	V _{DD} (V)	- 40°C		25°C			85°C		Unit
					Min.	Max.	Min.	Typ.	Max.	Min.	Max.	
High-Level Output Voltage		V _{OH}	I _{OUT} < 1 μA V _{IN} = V _{SS} , V _{DD}	5	4.95	—	4.95	5.00	—	4.95	—	V
				10	9.95	—	9.95	10.00	—	9.95	—	
				15	14.95	—	14.95	15.00	—	14.95	—	
Low-Level Output Voltage		V _{OL}	I _{OUT} < 1 μA V _{IN} = V _{SS} , V _{DD}	5	—	0.05	—	0.00	0.05	—	0.05	V
				10	—	0.05	—	0.00	0.05	—	0.05	
				15	—	0.05	—	0.00	0.05	—	0.05	
Output High Current		I _{OH}	V _{OH} = 4.6V	5	− 0.61	—	− 0.51	− 1.0	—	− 0.42	—	mA
			V _{OH} = 2.5V	5	− 2.50	—	− 2.10	− 4.0	—	− 1.70	—	
			V _{OH} = 9.5V	10	− 1.50	—	− 1.30	− 2.2	—	− 1.10	—	
			V _{OH} = 13.5V	15	− 4.00	—	− 3.40	− 9.0	—	− 2.80	—	
			V _{IN} = V _{SS} , V _{DD}									
Output Low Current		I _{OL}	V _{OL} = 0.4V	5	0.61	—	0.51	1.2	—	0.42	—	mA
			V _{OL} = 0.5V	10	1.50	—	1.30	3.2	—	1.10	—	
			V _{OL} = 1.5V	15	4.00	—	3.40	12.0	—	2.80	—	
			V _{IN} = V _{DD}									
Input High Voltage		V _{IH}	V _{OUT} = 0.5V	5	3.5	—	3.5	2.75	—	3.5	—	V
			V _{OUT} = 1.0V	10	7.0	—	7.0	5.50	—	7.0	—	
			V _{OUT} = 1.5V	15	11.0	—	11.0	8.25	—	11.0	—	
			I _{OUT} < 1 μA									
Input Low Voltage		V _{IL}	V _{OUT} = 4.5V	5	—	1.5	—	2.25	1.5	—	1.5	V
			V _{OUT} = 9.0V	10	—	3.0	—	4.50	3.0	—	3.0	
			V _{OUT} = 13.5V	15	—	4.0	—	6.75	4.0	—	4.0	
			I _{OUT} < 1 μA									
Input Current	"H" Level	I _{IH}	V _{IH} = 18V	18	—	0.1	—	10 ^{−5}	0.1	—	1.0	μA
	"L" Level	I _{IL}	V _{IL} = 0V	18	—	− 0.1	—	− 10 ^{−5}	− 0.1	—	− 1.0	
Quiescent Supply Current		I _{DD}	V _{IN} = V _{SS} , V _{DD} *	5	—	0.25	—	0.001	0.25	—	7.5	
				10	—	0.50	—	0.001	0.50	—	15.0	
				15	—	1.00	—	0.002	1.00	—	30.0	

* All valid input combinations.

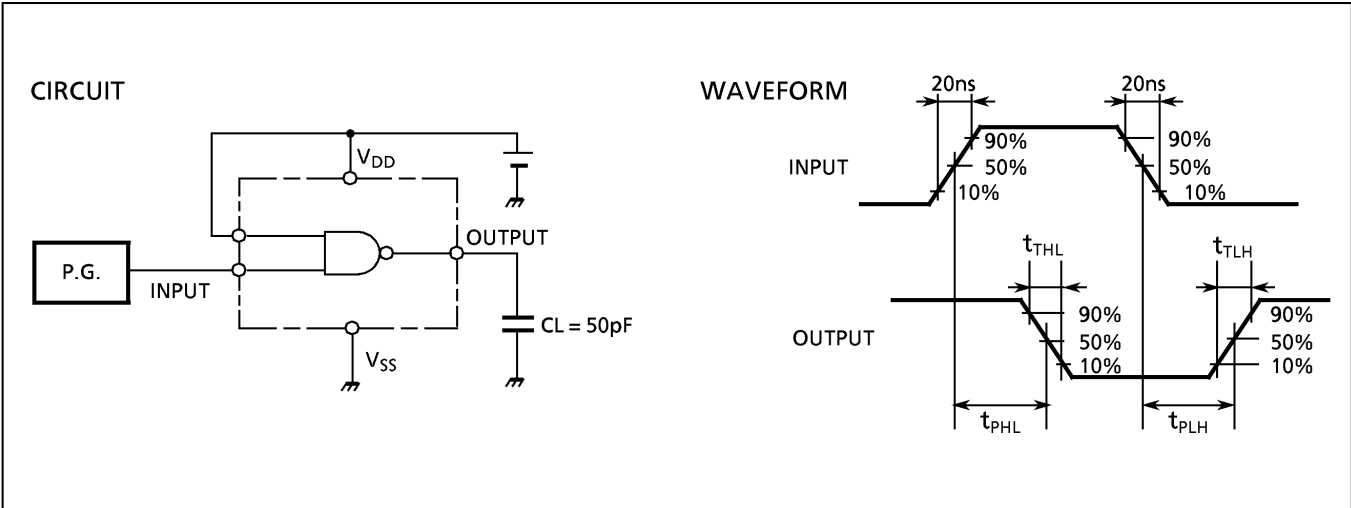
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DYNAMIC ELECTRICAL CHARACTERISTICS (Ta = 25°C, Vss = 0V, CL = 50pF)

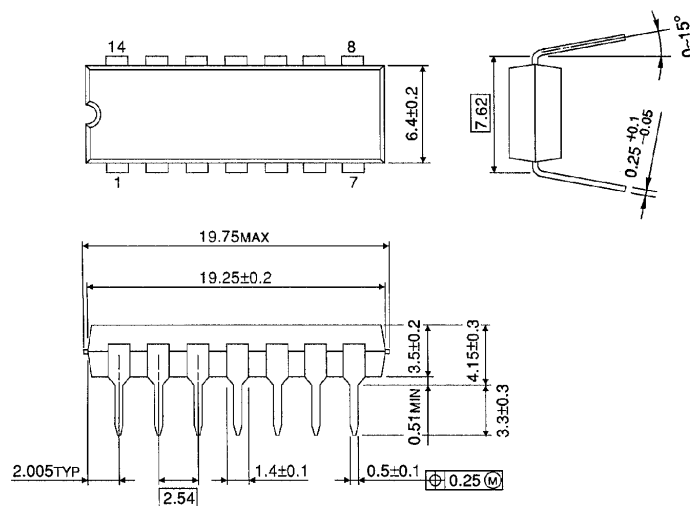
CHARACTERISTIC	SYMBOL	TEST CONDITION	VDD(V)	MIN.	TYP.	MAX.	UNIT
Output Transition Time	tTLH		5	—	70	200	ns
			10	—	35	100	
			15	—	30	80	
Output Transition Time	tTHL		5	—	70	200	
			10	—	35	100	
			15	—	30	80	
Propagation Delay Time	tPLH		5	—	65	200	
			10	—	30	100	
			15	—	25	80	
Propagation Delay Time	tPHL		5	—	65	200	
			10	—	30	100	
			15	—	25	80	
Input Capacitance	CIN			—	5	7.5	pF

CIRCUIT AND WAVEFORM FOR MEASUREMENT OF DYNAMIC CHARACTERISTICS



DIP 14PIN OUTLINE DRAWING (DIP14-P-300-2.54)

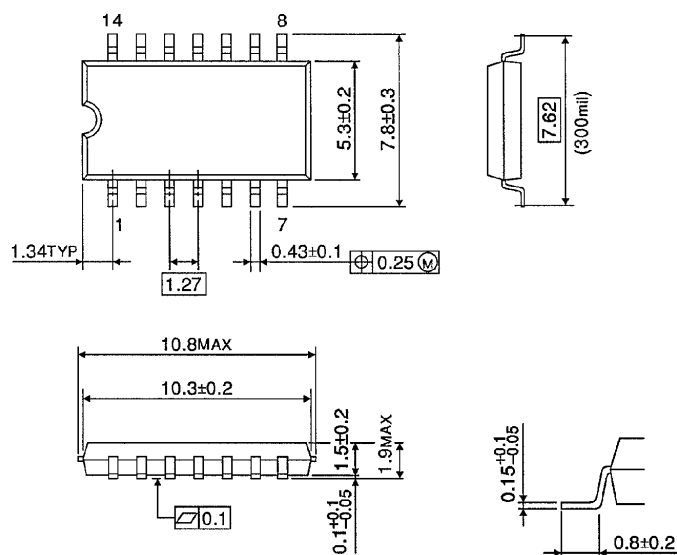
Unit in mm



Weight : 0.96g (Typ.)

SOP 14PIN (200mil BODY) OUTLINE DRAWING (SOP14-P-300-1.27)

Unit in mm



Weight : 0.18g (Typ.)

Unit in mm

Technical drawing of a 14-pin D-sub connector. The drawing includes a top view, a side view, and a detail of the contact pin.

Top View Dimensions:

- Pin 14 to Pin 8 (top row): 3.9 ± 0.1
- Pin 8 to Pin 7 (bottom row): 6.0 ± 0.2
- Pin 1 to Pin 7 (bottom row): 0.515 TYP
- Pin 1 to Pin 7 (top row): 0.515 TYP
- Pin 1 to Pin 7 (bottom row): 1.27
- Pin 1 to Pin 7 (top row): 0.42 ± 0.07
- Pin 1 to Pin 7 (bottom row): 0.25

Side View Dimensions:

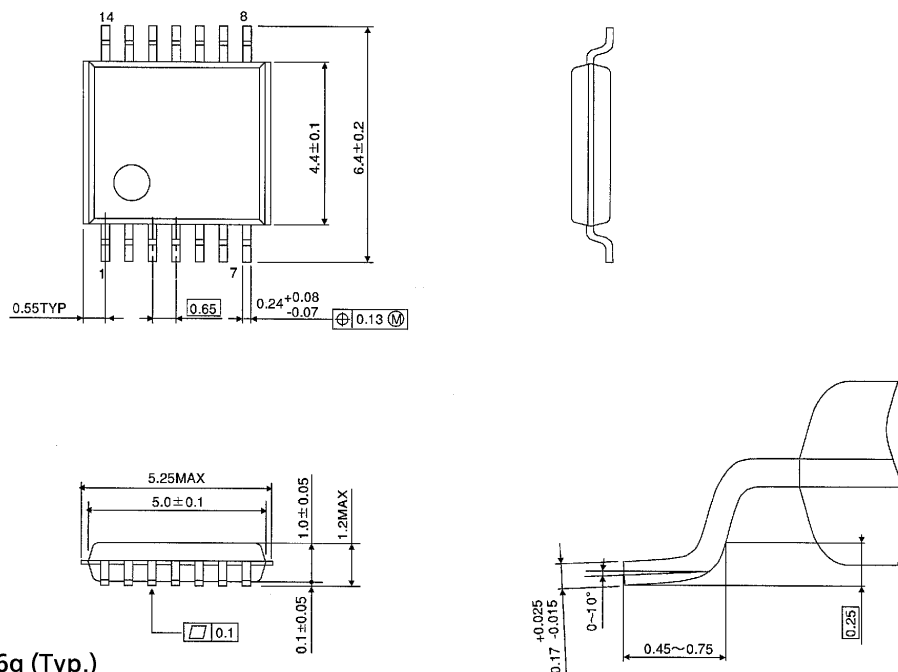
- Pin 14 to Pin 8 (top row): 8.65 ± 0.1
- Pin 8 to Pin 7 (bottom row): 1.375 ± 0.2
- Pin 1 to Pin 7 (bottom row): 0.175 ± 0.075
- Pin 1 to Pin 7 (top row): 1.75 MAX

Detail Dimensions:

- Pin 1 to Pin 7 (bottom row): 0.1
- Pin 1 to Pin 7 (top row): 0.15 ± 0.05
- Pin 1 to Pin 7 (bottom row): 0.7 ± 0.3
- Pin 1 to Pin 7 (top row): 45°

Weight : 0.12g (Typ.)

Unit in mm



Weight : 0.06g (Typ.)